Detecting LLM Hallucinations with Uncertainty Fingerprinting

A Python implementation tutorial

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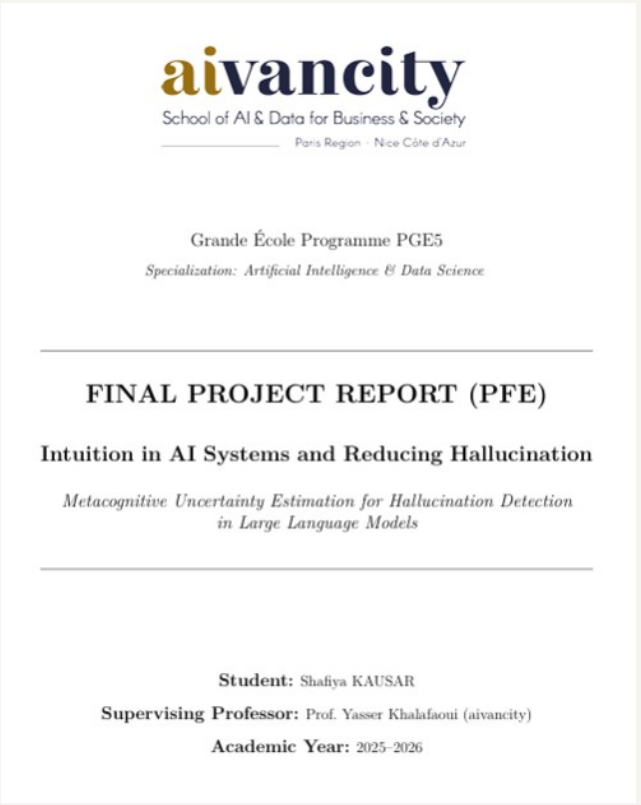
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Specialization: Artificial Intelligence & Data Science

FINAL PROJECT REPORT (PFE)

Intuition in AI Systems and Reducing Hallucination

Metacognitive Uncertainty Estimation for Hallucination Detection in Large Language Models

Student: Shafiya KAUSAR

Supervising Professor: Prof. Yasser Khalafaoui (aivancity)

Academic Year: 2025–2026

Sincere thanks: Dr. Yasser Khalafaoui, Master thesis supervisor

Learning objectives

IMPORTANT: go to the GitHub repo and run the setup check notebook!

By the end of this hands-on workshop, you will be able to:

- Explain what an AI hallucination is, the main kinds, and why an AI can sound sure when it is wrong
- Write code for four simple checks that show when an AI is unsure, and see which one works best
- Spot patterns that show which answers are likely hallucinated
- Decide when an AI should answer, double-check, or say “I don’t know”
- Try a live app to play with hallucination detection!

Scan to open the repo

Setup: do this before we start (5 min)




```
git clone \
https://github.com/Shafiya0101/pycones2026-uncertainty-workshop.git
cd pycones2026-uncertainty-workshop
python -m venv .venv
source .venv/bin/activate   # Windows: .venv\Scripts\activate
pip install -r requirements.txt
jupyter lab
```

THEN

Open `00_setup_check` and run all cells

GOOD TO KNOW

CPU only · no API key · Colab works too

Workshop roadmap · 90 min

End goal: a working detector that tells you when to trust an LLM answer, and what kind of failure it is.

a · 10 min	**All about hallucinations**	What are hallucinations and its types	**Live demo**
b · 20 min	**Uncertainty signals**	What clues can be used to detect hallucinations	**Notebook 01**
c · 15 min	**Uncertainty fingerprinting**	Detecting patterns in hallucinations	**Notebook 02**
d · 15 min	**Adaptive policy**	When should the AI answer, check, or refuse to answer if there is hallucination?	**Notebook 03**
e · 25 min	**Experiments & live demo**	Try our hallucination analysis app!	**App**
f · 5 min	**Remarks and conclusion**	Limitations and future work	

Workshop steps

Detect → Combine → Decide → Stress-test

SETUP — 2 min

Setup check

Check that your laptop is ready

`00_setup_check`

CHALLENGE 1 — 3 min

Spot the hallucination

Can you spot the AI's made-up answer?

`Raise your hand`

CHALLENGE 2 — 12 min

Uncertainty signals

Four clues that show when an AI is guessing

`01_uncertainty_signals`

CHALLENGE 3 — 12 min

Fingerprinting

Sort answers into safe and risky groups

`02_fingerprinting`

CHALLENGE 4 — 12 min

Adaptive policy

When should the AI answer, check, or refuse?

`03_adaptive_policy`

CHALLENGE 5 — 15 min

App based experimentation

Find answers that fool the checks

Basics: five words to know

Word	Plain meaning	Example	Used in
LLM	**Predicts the next word, again and again**	“The capital of France is …” → “Paris”	*Everywhere*
Uncertainty signal	A clue that shows how sure the AI is	10 different answers = the AI is guessing	*Notebook 01*
Fingerprinting	Putting an answer in a safe or risky box, using two clues	Same answers = safe box. Different answers = risky box	Notebook 02
Greedy answer	**Most likely answer: what users see**	Temperature 0 = same answer every time	*All notebooks*
Samples	**Random extra answers from the same model**	10 samples per question	*Notebook 01*
Ground truth	**Correct answer from the dataset**	“Stilwell” for “Vinegar Joe”	*Evaluation*

What is a hallucination?

Fluent, confident output of an LLM that is **not supported** by the world, the input, or the model's own reasoning.

FACTUALITY

Wrong about the world

Invents facts, or gets a real fact wrong.

"The Eiffel Tower opened in 1901."

FAITHFULNESS

Wrong about the input or itself

Ignores the instruction, contradicts the source, or contradicts itself.

"Step 2 says 12; the answer says 15."

Types of hallucinations

Family	Type	What happens	Example
Factuality	**Fabrication**	Invents facts or entities	*Cites a paper that was never written*
Factuality	**Factual inconsistency**	Gets a real fact wrong	*"The Eiffel Tower opened in 1901."*
Faithfulness	**Instruction inconsistency**	Ignores what was asked	*Asked to translate, it answers instead*
Faithfulness	**Context inconsistency**	Contradicts its source	*Summary says sales rose; the report says they fell*
Faithfulness	**Logical inconsistency**	Reasoning conflicts with itself	*Step 2 says 12; the answer says 15*

Taxonomy after Huang et al., "A Survey on Hallucination in Large Language Models" (2023)

A confident fabrication

Illustrative example

PROMPT	“Summarize the 2019 paper *‘Neural Entropy Gates for Language Models’* by Hartmann et al.”
MODEL OUTPUT	“Hartmann et al. (2019) introduced entropy gates in NeurIPS 2019, reducing perplexity by 12% on Penn Treebank by dynamically gating hidden-state updates…”
REALITY	**The paper doesn’t exist.** Authors, venue and numbers are all invented, yet it reads like a real citation.
REAL-WORLD CASE	*Mata v. Avianca* (2023): a New York lawyer filed ChatGPT-invented case citations, including “Varghese v. China Southern Airlines”. The model even said the cases were real.

Confident fabrications are not rare

5–29%

When the AI gave the exact same answer all 10 times, it was still wrong in 5–29% of cases (Mistral-7B)

6.5–57%

Among answers where the AI seemed confident, between 6.5% and 57% were wrong, depending on the model and dataset.

Source: study paper (Section V-D, Table VI) and workshop README

Can you detect the hallucination?

⏱️ 2 min

Four answers from an LLM. **Raise your hand** for each one you think is wrong.

A

Who wrote Frankenstein?

"Mary Shelley. It was first published anonymously in 1818."

B

First woman to win the Nobel Prize in Literature?

"Pearl S. Buck, who received the prize in 1938 for her novels about rural China."

C

Which element has the symbol W?

"Tungsten. The symbol comes from its German name, Wolfram."

D

Which paper introduced the Transformer?

"'Attention Is All You Need' by Vaswani et al., presented at ICML 2017."

The reveal: 2 of 4 were hallucinations

Same fluency, same confidence. The errors hide **right next to true facts.**

A ✓ **Correct**

“Mary Shelley. It was first published anonymously in 1818.”

B × **Hallucination**

“Pearl S. Buck, who received the prize in 1938…”

Actually: Selma Lagerlöf, in 1909. Buck won in 1938, but wasn’t first.

C ✓ **Correct**

“Tungsten. The symbol comes from its German name, Wolfram.”

D × **Hallucination**

“…by Vaswani et al., presented at ICML 2017.”

Actually: NeurIPS 2017 (then called NIPS), not ICML.

Why are hallucinations hard to detect?

01

Good writing is not proof

A wrong answer reads just as smoothly as a right one. You can’t tell by the style.

02

It sounds sure even when it’s wrong

The AI doesn’t hesitate or hedge when it makes a mistake. It sounds just as certain.

03

Nobody checks the answer

When people use the AI for real, there is no answer sheet to check against. We have to judge from the AI’s own behaviour.

A hallucination looks exactly like a correct answer!

Why are hallucinations hard to detect?

04

Repeating is not the same as right

Ask the AI ten times and it may give the same wrong answer ten times.

05

Every check has blind spots

Entropy, logits, consistency and NLI each miss different some kinds of mistakes.

06

Checking costs time and money

Asking ten times and checking every answer takes much more time and computing than asking once.

Takeaway

No single check is enough. We combine several, and we study the cases where every check says “fine” but the answer is still wrong.

What the research says (1/2)

SINGLE SIGNALS HAVE BLIND SPOTS

NATURE 2024

Farquhar et al.

Semantic entropy targets confabulations only. The authors exclude systematic, learned errors.

FINDINGS OF EMNLP 2025

Simhi et al., “Trust Me, I’m Wrong”

Models hallucinate with high certainty, even when they know the answer. Existing mitigations struggle with these cases.

What the research says (2/2)

COMBINING SIGNALS HELPS

UQLM · NeurIPS 2025

Ensemble beats the best single scorer in 18 of 24 AUROC cases. Python library.

HaluNet · 2025

Fuses token likelihood, entropy and hidden states in one network.

BEACON · arXiv 2026

Combines NLI entropy, embeddings, CoT consistency and paraphrase stability.

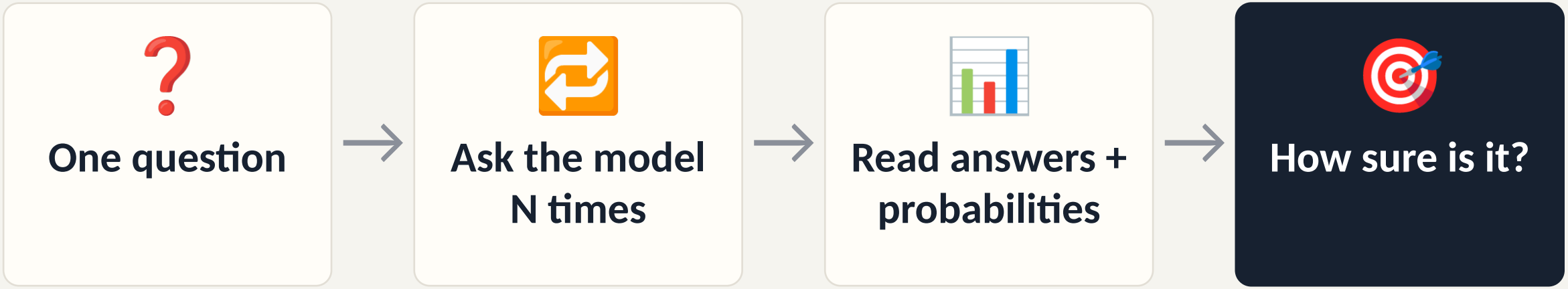
Chauhan et al. · 2026

Supervised ensembles beat the best single scorer in 30 of 32 settings.

Our step Others merge signals into one score. We read the **pattern** to classify the type of hallucination, and act on it.

What are uncertainty signals?

An uncertainty signal is a clue in how the AI behaves that shows how sure it is. We don’t need to know the right answer.



Think of a person who is unsure

Spotting an unsure person...	...spotting an unsure AI
Gives a different answer every time you ask	→ **Self-consistency** · do the answers match?
Keeps changing what they actually mean	→ **Semantic entropy** · how many different meanings?
Hesitates and stumbles over words	→ **Logit confidence** · how unsure was it of each word?
Contradicts what they said earlier	→ **Contradiction** · do the other answers disagree with the one shown?

The four uncertainty signals (1/2)

01 · AGREEMENT

Self-consistency

Does it give the same answer every time?

How: Sample 10 answers, measure agreement with the biggest group.

02 · MEANING

Semantic entropy

Do the answers mean different things?

How: Cluster samples by meaning, then compute entropy over clusters.

The four uncertainty signals (2/2)

03 · PROBABILITY

Logit confidence (LP)

How Sure is the model sure of each token (word) it generates?

How: Aggregate the token log-probabilities of the shown answer.

04 · LOGIC

Contradiction

If the generated answers are contradicting with each other?

How: An NLI model checks each sample against the greedy answer.

Datasets used

3 datasets · 5 settings × 200 questions

Dataset	Description	Project page
TriviaQA (Joshi et al., 2017)	General-knowledge quiz questions · ~95k	nlp.cs.washington.edu/triviaqa
Natural Questions (Kwiatkowski et al., 2019)	Real Google search questions · ~307k	ai.google.com/research/NaturalQuestions
HotpotQA (Yang et al., 2018)	Questions needing two facts combined · ~113k	hotpotqa.github.io

How we ran the study (1/2)

5 settings × 200 questions

MODELS	**Three AI models:** Mistral-7B (main), Qwen2-7B and Qwen2-1.5B. Small, compressed versions that run on a laptop.
DATA	**Three question sets:** TriviaQA, Natural Questions and HotpotQA. 200 randomly chosen questions from each.
GENERATION	**Ten answers per question:** each question we generated 10 answers

How we ran the study (2/2)

5 settings × 200 questions

SIGNALS	**THE CHECKS:** All four: matching answers, matching meanings, the AI's word confidence, and contradictions
LABELS	**MARKING ANSWERS RIGHT OR WRONG:** We use label containment, word - matching rule, secondly, LLM as a judge, which in this case is Claude Haiku.
METRICS	**THE SCORE:** AUROC: how well a check spots wrong answers. 50% = guessing, 100% = perfect. .

Key terms · Notebook 01

Term	What it means	Example	How to read it
Log-prob. (LP)	**How sure the model is of each word it writes**	Low confidence per word = high LP score	*Higher = less sure*
Self-consist. (SC)	**Share of the 10 samples that differ in wording**	10 of 10 identical = 0%	*Higher = less sure*
Semantic entropy (SE)	**Spread of the 10 samples over different meanings**	“Paris” and “the city of Paris” = 1 meaning	*Higher = less sure*
NLI contradiction	**Chance a sample contradicts the greedy answer**	80% = most samples disagree with it	*Higher = less sure*
AUROC	**Chance a wrong answer looks less sure than a right one**	50% = coin flip, 100% = perfect	*Higher = better detector*

NLI = natural language inference. AUROC = area under the ROC curve.

Hands-on · Notebook 01

⏱ ≈ 12 min

Greedy answer + 10 samples per question, precomputed. **Higher score = more uncertain.**

Implementations

STEP 1 Self-consistency
STEP 2 Semantic entropy
STEP 3 Logit confidence
STEP 4 Contradiction
Step 5 Combine all four signals

Which signal is best to detect hallucination?

AUROC: 50% = guessing, 100% = perfect. Predict first, then run.

- Bonus: Done early? Change line to load_setting("mistral_hotpotqa", “qwen2_1_5b_triviaqa”, “qwen2_7b_triviaqa”) and re-run. Is any signal still the best?

Run the code, check the signal values, check corresponding AUROCs and determine which signal works best.

Use it in your own work (1/2)

The core idea: ask the AI the same question several times and see whether the answers agree.

"When did the Eiffel Tower open?" × 5

1889 · 1889 · 1889 · 1889 · 1889

✔ **They agree: more likely right**

"When did the Eiffel Tower open?" × 5

1889 · 1901 · 1889 · 1912 · 1895

⚠️ **They disagree: don't trust it yet**

Agreement means "more likely right", not "guaranteed right".

Use it in your own work (2/2)

STEP 1

Ask more than once

Send each question 5–10 times; keep every answer.

STEP 2

Compare answers

The notebook's code checks if they match in words and meaning.

STEP 3

Test on your topic

Check 100–200 answers by hand: which check works best?

STEP 4

Decide what to do

If unsure: say "I don't know", look it up, or ask a person.

📓 **Take the notebook home: it gives you**

Working, tested code
Swap in your own questions and answers.

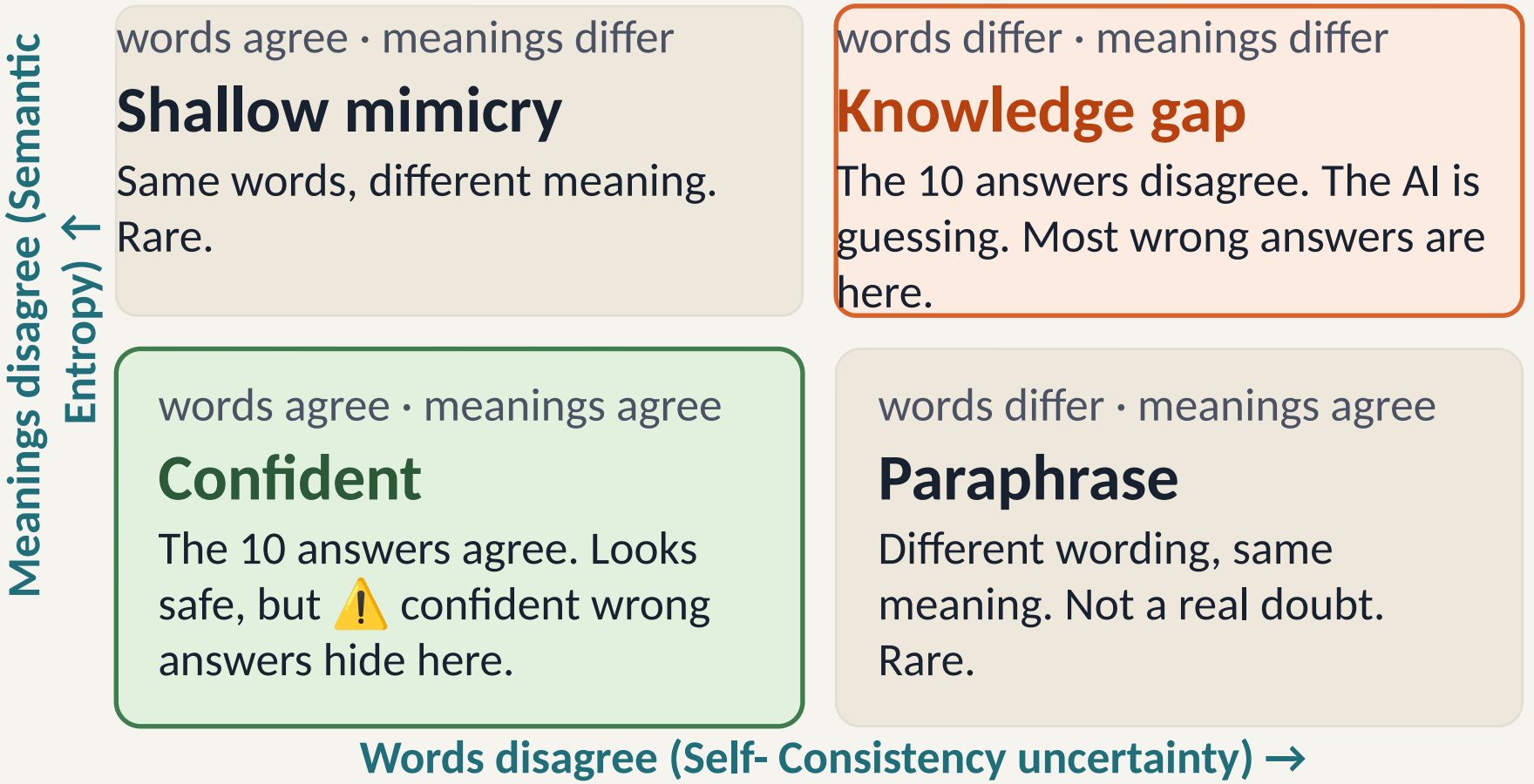
A ready-made scoreboard
One line shows which check works best.

Runs on any laptop
No GPU or paid service needed.

Fits: customer chatbots · ask-your-documents tools · sorting data with AI · testing a new model

Uncertainty fingerprinting

A fingerprint shows where an answer lands on two checks: do the 10 answers use the same words, and do they mean the same thing? That sorts every answer into one of four boxes.



What the four boxes show

REAL RESULTS · MISTRAL-7B ON TRIVIAQA · 200 QUESTIONS

Confident
(answers agree) / 92 questions· **6.5% wrong**

Shallow
/ 10 questions · too few to judge

Paraphrasing
/ 11 questions · too few to judge

Knowledge gap
(answers disagree) / 87 questions · **42.5% wrong**

HOW TO READ IT

- **The red flag:** when the AI's answers disagree, it is wrong far more often (42.5% versus 6.5%). Knowledge gap had the most errors in every setting we tested.
- **A simple rule:** if the answers disagree, don't answer. We don't need an answer key for this.
- **The catch:** a plain score cut-off does about as well, so the boxes are not magic. More on this later.

```
df = add_signals(load_setting("mistral_triviaqa"))
```

Now lets change the model to Qwen and see the effect on the fingerprints!

Key terms · Notebook 02

Term	What it means	Example	In our data
Fingerprint	**The cell of the SC × SE grid an answer falls in**	Confident, Gap-like, Shallow, Surface-varying	*4 cells*
Median threshold	**Cut that splits answers into low and high**	Half the answers fall below it	*No labels needed*
Confident region	**Both signals low: the samples agree**	Mistral on TriviaQA	*92 of 200 · 6.5% wrong*
Gap-like region	**Both signals high: the samples disagree**	Mistral on TriviaQA	*87 of 200 · 42.5% wrong*
Unanimous answer	**All 10 samples identical**	Still wrong sometimes	*74 answers · 5 wrong*

SC = self-consistency, SE = semantic entropy. Figures: Mistral-7B on TriviaQA.

Hands-on · Notebook 02

⏱ ≈ 12 min

See how different models have different fingerprints. **No code to write.**

STEP 1 · RUN

See Mistral-7B's fingerprint

Do: run the first cell

Look for: which box has most wrong answers?

```
show_fingerprint("mistral_triviaqa")
```

→

STEP 2 · YOUR TURN

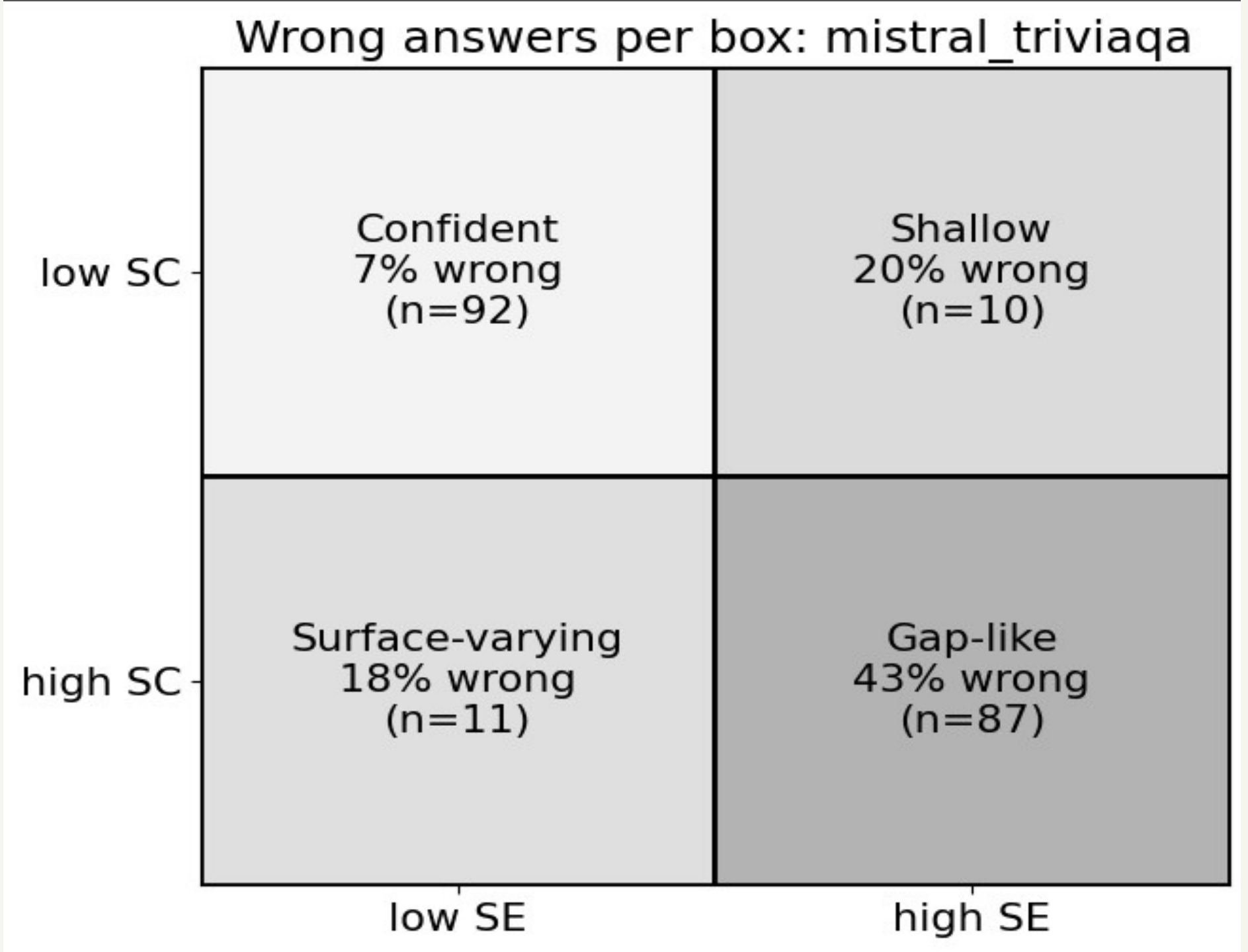
Switch to the Qwen model

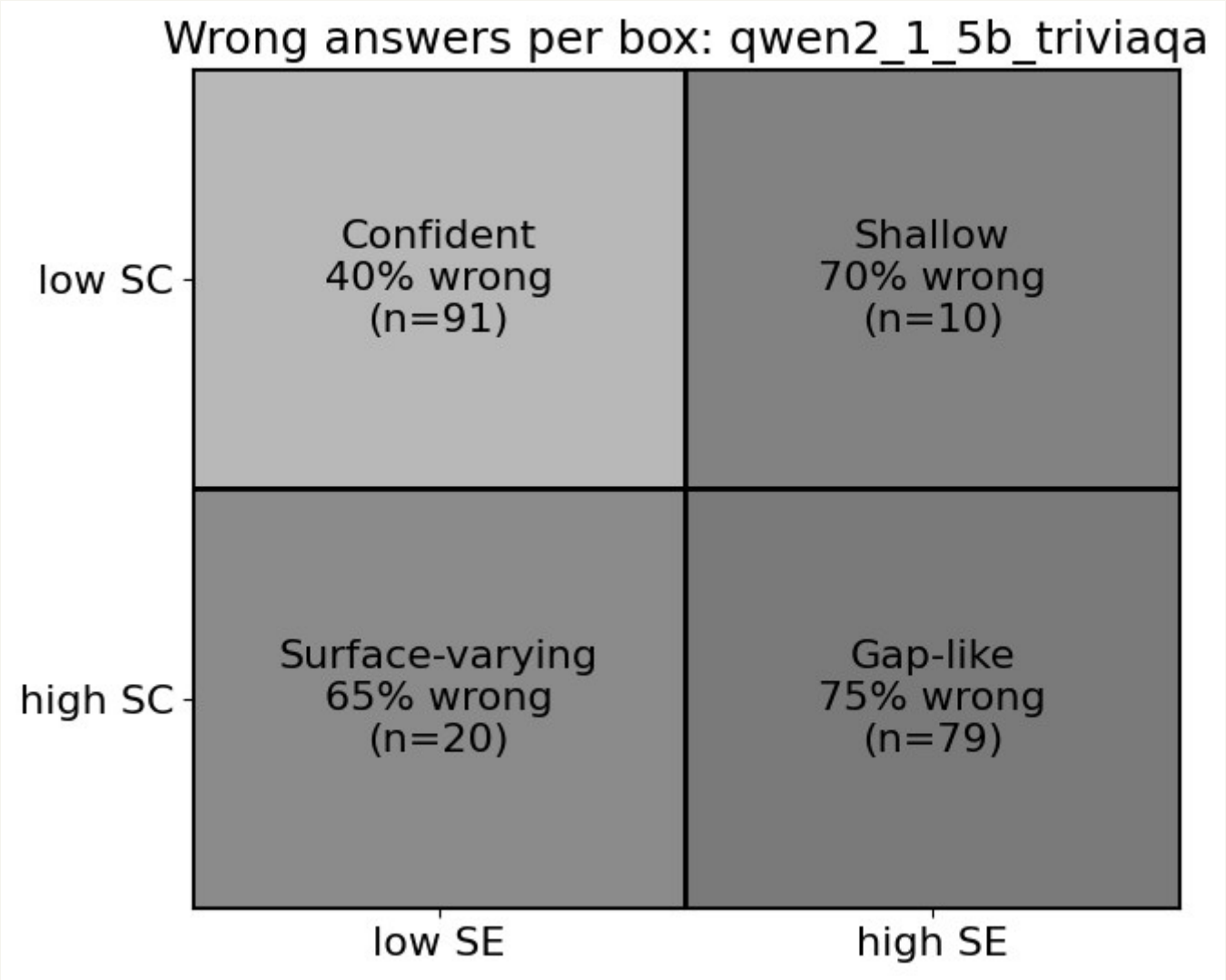
Do: change the name in cell 2, then run it

Look for: what changed in the "Confident" box?

```
show_fingerprint("qwen2_1_5b_triviaqa")
```

Done early? Try `qwen2_7b_triviaqa` · `mistral_nq` · `mistral_hotpotqa`


Wrong answers per box: mistral_triviaqa
Confident
7% wrong
(n=92)
Shallow
20% wrong
(n=10)
Surface-varying
18% wrong
(n=11)
Gap-like
43% wrong
(n=87)
low SC
high SC
low SE
high SE


Wrong answers per box: qwen2_1_5b_triviaqa
Confident
40% wrong
(n=91)
Shallow
70% wrong
(n=10)
Surface-varying
65% wrong
(n=20)
Gap-like
75% wrong
(n=79)
low SC
high SC
low SE
high SE

Discuss

⏱ 3 min

Look at the table of **unanimous-but-wrong** answers from Step 4.

1. Which model's "Confident" box can you trust more? How can you tell?

2. Why might a smaller model be wrong even when its answers agree?

3. Would you use the same cut-offs for every model? Why not?

Bonus: Try mistral_nq or mistral_hotpotqa. Does the "Confident" box get more or less reliable?

What notebook 02 shows

Most answers land in two boxes

Same words usually means same meaning, so answers are mostly Confident or Knowledge gap.

Disagreement is the warning sign

When the AI's answers disagree, it is wrong far more often. We can spot this without an answer key.

Confident does not mean correct

Some wrong answers sit in the Confident box. Only an outside source, like a document or a person, can catch them.

Every model is different

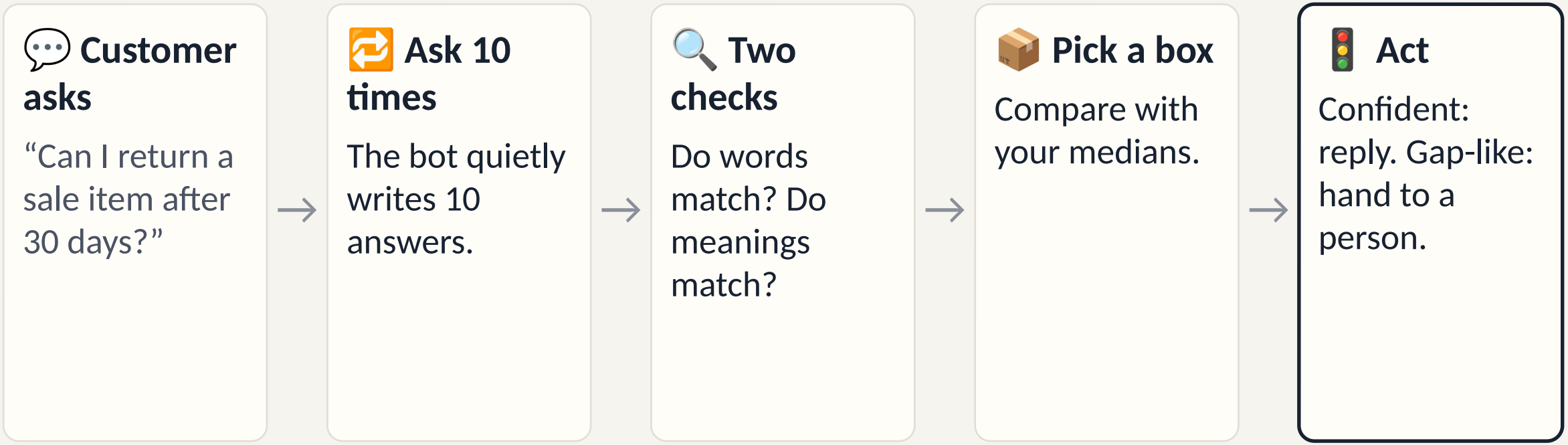
The smaller Qwen model was wrong about 40% of the time even in its Confident box, versus about 6.5% for Mistral.

Next Turn the boxes into a rule. When should the AI answer, double-check, or say "I don't know"?

Use fingerprints in your own work (1/2)

Our message: you can sort AI answers by risk **without knowing the right answer.** But "Confident" still doesn't mean "correct".

EXAMPLE · A SUPPORT CHATBOT



NEXT!

Adaptive policy: A rule for when the AI answers, double-checks, or says “I don’t know”

Measuring how unsure the AI is isn’t enough. Use it to decide what to do with each question.

← Low uncertainty

High uncertainty →

✅ Answer

the AI is sure, so reply straight away.

🔍 Verify

look it up or ask a bigger model before replying.

✋ Abstain

don’t risk a wrong answer.

Rule = (U_lp + U_sc + U_se + U_cont) / 4

Accuracy and coverage curves

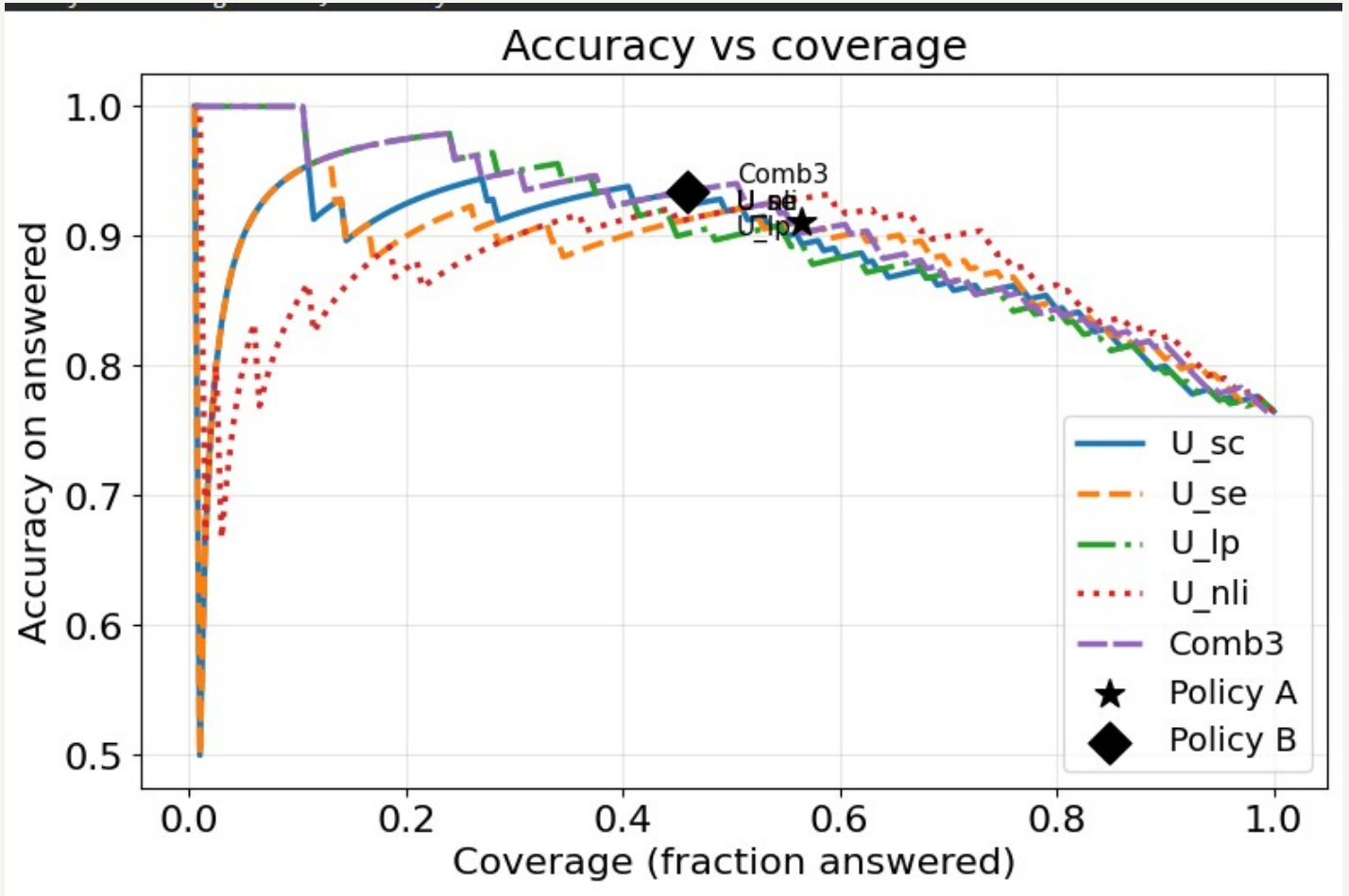
The policy you build lands on one point on this curve. It gives two numbers: how many questions got answered (coverage) and how many of those were right (accuracy).

COVERAGE

how many it answers. Out of 100 questions, how many does the AI actually answer?

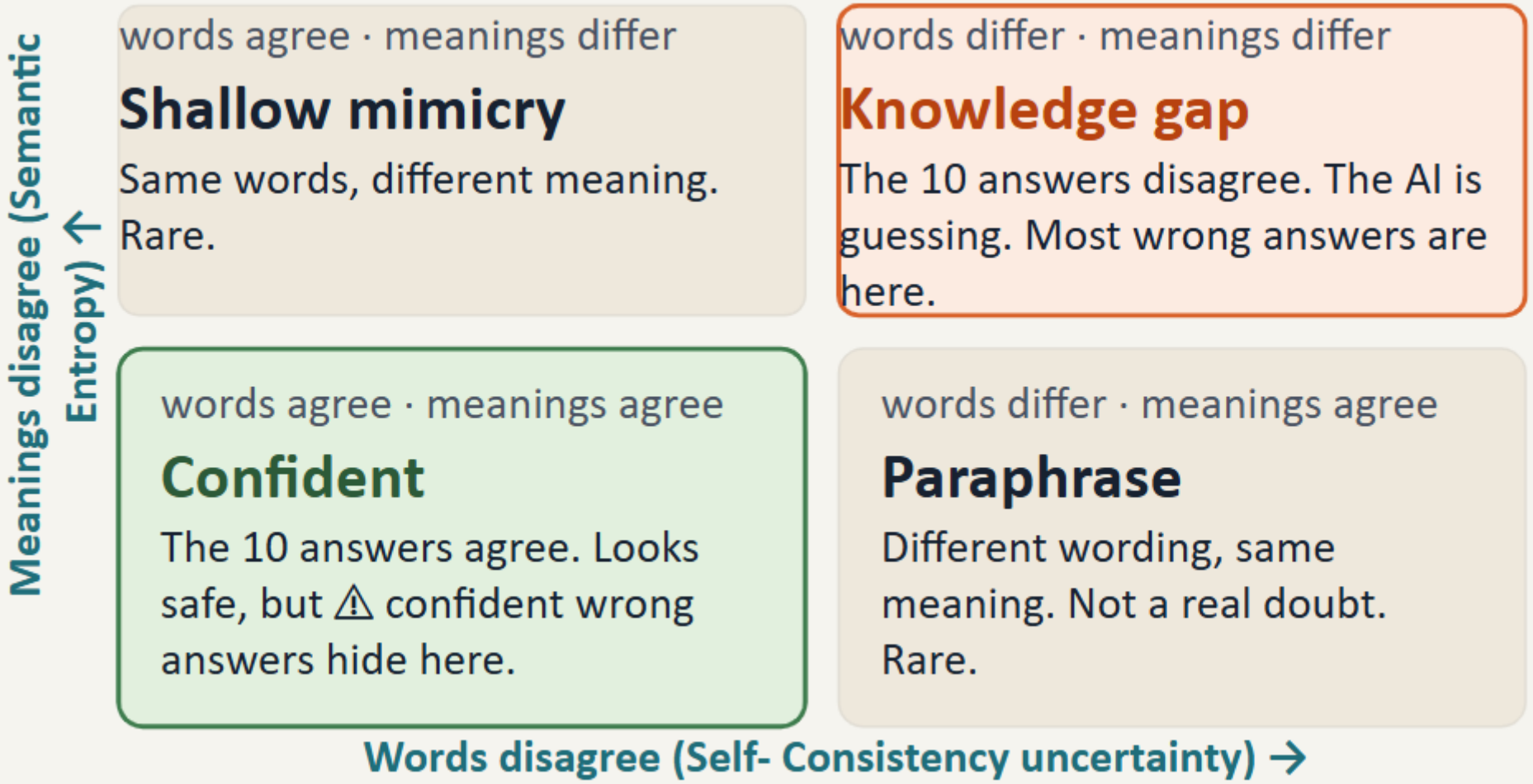
ACCURACY

how often it's right. Of the questions it answered, how many were correct?



Policy A: answer everything except the Gap-like box.
Policy B: answer only the Confident box

Reminder!


words agree · meanings differ
Shallow mimicry
Same words, different meaning. Rare.
words differ · meanings differ
Knowledge gap
The 10 answers disagree. The AI is guessing. Most wrong answers are here.
words agree · meanings agree
Confident
The 10 answers agree. Looks safe, but ⚠ confident wrong answers hide here.
words differ · meanings agree
Paraphrase
Different wording, same meaning. Not a real doubt. Rare.
Meanings disagree (Semantic Entropy) ↑
Words disagree (Self- Consistency uncertainty) →

Key terms · Notebook 03

Term	What it means	Example	In our data
Coverage	**Share of questions the system answers**	Answers 56 of 100 questions = 56%	*Policy A: 56.5%*
Accuracy on answered	**Share of answered questions that are right**	Only counts the answers given	*Policy A: 91.2%*
Abstain	**Say “not sure” instead of answering**	Used on Gap-like answers	*Policy A*
Verify	**Send to a costlier extra check**	Retrieval or a human	*Extra cost*
Matched coverage	**Compare policies that answer the same share**	Fingerprint vs plain threshold	*No clear gap*

Policy A = abstain on Gap-like answers. Figures: Mistral-7B on TriviaQA.

Hands-on · Notebook 03

⏱️ ≈ 12 min

Be the AI's manager: decide when it **answers**, **double-checks**, or says **"I don't know".**

STEP 1 · RUN
Observe the Accuracy-Coverage curve

STEP 2 · RUN
Sort questions into three groups: answer, double-check, "I don't know".

STEP 3 · Experiment

Change the thresholds

```
q_answer, q_abstain = 0.45, 0.80
```

q_answer and q_abstain are two dials that decide how careful the AI should be.

STEP 4 · Analyse
Do you think accuracy is better in fingerprint?

	q_answer	q_abstain	coverage	accuracy	cost_per_question
0	0.3	0.8	0.8	0.919	2.0
1	0.3	0.9	0.9	0.917	2.2
2	0.3	1.0	1.0	0.915	2.4
3	0.5	0.8	0.8	0.919	1.6
4	0.5	0.9	0.9	0.917	1.8
5	0.5	1.0	1.0	0.915	2.0
6	0.7	0.8	0.8	0.900	1.2
7	0.7	0.9	0.9	0.900	1.4
8	0.7	1.0	1.0	0.900	1.6

	q_answer	q_abstain	coverage	accuracy	cost_per_question
0	0.3	0.8	0.8	0.919	2.0
1	0.3	0.9	0.9	0.917	2.2
2	0.3	1.0	1.0	0.915	2.4
3	0.5	0.8	0.8	0.919	1.6
4	0.5	0.9	0.9	0.917	1.8
5	0.5	1.0	1.0	0.915	2.0
6	0.7	0.8	0.8	0.900	1.2
7	0.7	0.9	0.9	0.900	1.4
8	0.7	1.0	1.0	0.900	1.6

Assistance understanding notebook 3

Imagine a LLM that answers questions, and you want to know when to believe it.

Signals: you ask the LLM the same question 10 times. If it says the same thing every time, it is probably sure. If it keeps changing its answer, it is guessing. These are the clues.

Score: you put all the clues together into one number. A small number means “the LLM is sure.” A big number means “the LLM is guessing.”

Fingerprint: you sort the questions into boxes. One box is “sure,” one is “guessing,” and so on. The boxes tell you why the LLM is unsure.

Dials (q_answer, q_abstain): these decide what the LLM does:

- Very sure: it answers right away.
- A bit unsure: it checks first.
- Very unsure: it says “I don’t know.”

Take Away

⏱ 2 min

Fingerprinting of hallucination patterns isn’t enough, we need to use it to take decisions how our LLM behaves.

What we learned from notebook 03

Being right costs answers

The more careful the AI is, the fewer questions it answers. Choose the balance on purpose.

Double-checking is the middle path

Spend extra effort only on the questions the AI is unsure about.

The boxes explain

Even if the accuracy is similar, they tell you why the AI held back.

Next our results across 3 models and 3 datasets, then a live demo.

Use a policy in your own work (1/2)

Our message: decide in advance what your AI does when it is sure, unsure, or very unsure.

EXAMPLE · A COMPANY HR ASSISTANT

✅ **Sure:** “How many days of leave do I have?” Answer right away.

🔍 **Unsure:** Look it up in the staff handbook first, then answer and show where it came from.

✋ **Very unsure:** Say “I’m not sure. Here’s who to ask in HR.”

Use a policy in your own work (2/2)

THREE DECISIONS TO MAKE BEFORE LAUNCH

1 · How much can you skip?

Set a minimum: for example, answer at least 60% of questions.

2 · What does “verify” cost?

Time, money, a bigger model or a person. Put real numbers on it.

3 · Which mistakes hurt?

The higher the stakes, the more you should verify or abstain.

📓 **Notebook 03** Route `my_actions` → score `evaluate_actions` → check it’s worth it `matched_coverage_comparison`

What we found

3 models · 3 datasets

0.63–0.82

How well the checks spot wrong answers (0.5 = guessing, 1 = perfect). Mixing checks was not clearly better.

0.86–0.93

How closely our two main checks agree (1 = identical). They carry almost the same information.

91.2%

Right while answering 56.5% of questions, with no answer key. (93.9% used the answer key, so it can't run for real.)

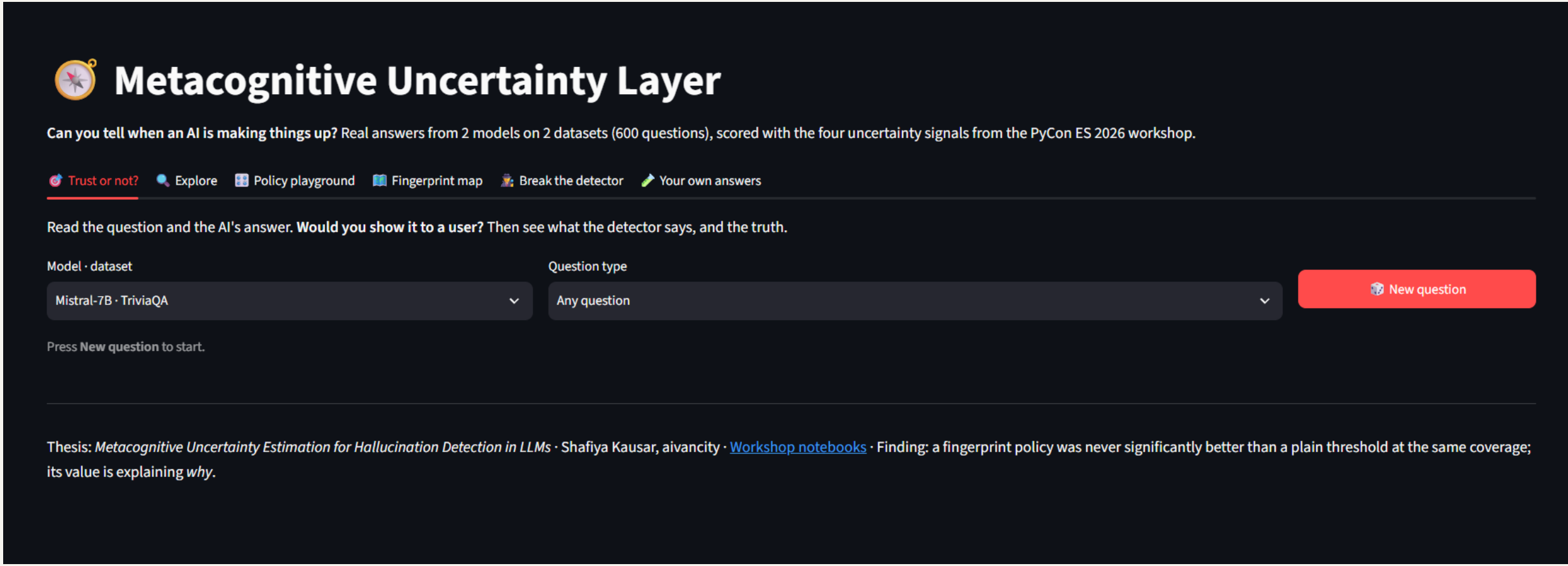
5–29%

Even when all 10 answers matched, this share was still wrong (Mistral-7B).

With 200 questions per setting, we found no clear difference between the box-based rule and a plain cut-off.

Finally! Playing with Hallucinations with our app

🔗 https://pycon2026-uncertainty-workshop.streamlit.app/



How to use the app?

1 **Section 1: Trust or not?**
Read a real question and the AI's answer, click "I'd trust it" or "I wouldn't", then see the truth, the signals and the detector's verdict.

Explore various possible hallucinations!

2 **Explore fingerprints**

Search or filter all 200 questions (confident fabrications, Gap-like, graders disagree…). For each one you see the answer, both grades, all 4 signals ("more uncertain than X% of questions") and the 10 samples grouped by meaning.

3 **Policy playground**
You're now the LLM's manager! For each question, the AI can answer straight away, get or say 'I don't know.' You decide what happens!

4 **Fingerprint map for models-dataset pair**

5 **Test your LLM!**

Help with using the Policy playground

- The left slider is **Answer cut off**: answer the safest X% directly. The right slider is **Abstain cut-off** : above this point, say ‘I don’t know.’ Everything in between gets double-checked.
- Drag both sliders to 1.0. This is a normal chatbot: it answers everything. 76.5% right, and **47 wrong answers reach users** out of 200.
- **Return to 0.45 and 0.80.** Now only **13** wrong answers get through, and accuracy is about 92%. **40 questions got ‘I don’t know’**, and **18 of those were actually right**. We threw away good answers.
- How to minimize this?

Left	Right	What it means
1.0	1.0	Answer everything: a normal chatbot
0.6	0.6	No verify zone: answer 60%, say “I don’t know” to 40%
0.2	1.0	Never abstain: answer 20%, verify the other 80%
0.0	0.0	Say “I don’t know” to everything

Grader setting chooses how the app decides whether each AI answer is **right or wrong**.

`label_containment` : **the word-matching rule** (“Word-matching rule” in the app)

- An answer counts as **correct** if it **contains** one of the accepted answers from the dataset.
- It’s simple and fast, but literal. It can be too generous, giving credit to a wrong answer that merely *mentions* the right word. It can also be too strict, marking a correct answer wrong because it uses different words.

`label_judge` : **the AI judge** (“AI judge” in the app)

- A separate LLM (Claude Haiku 5.5, per your evaluation setup) reads the question, the answer and the accepted answers, and decides whether the answer is correct.
- It’s better at meaning and synonyms, but it’s an AI too, so it can also make mistakes.

Break the detector

The detector said “fine”, but the answer was marked wrong. **Whose fault is it?**

🤖 Fabrication

The model really is wrong.

Says “Sydney” for the capital of Australia.

📏 Answer key too strict

Right answer, but the key wants exact words.

Says “physician”; the key says “doctor”.

🏷️ Grading mistake

The label itself is wrong.

A wrong answer gets credit for mentioning the right word.

🤷 Ambiguous

Reasonable people would disagree.

“Who invented the telephone?” Bell or Meucci?

Try it in the app

Scan to open the app



Open the app (Scan the QR code) → tab 🕵️ **Break the detector**.

Work in groups of 3–4. It takes about 7 minutes, and nobody needs to write code.

Tip: Step 2 lets you pick a cause for each case, and you can download your verdicts.

A fifth check like the other four, or something **completely different**?

Help for Break the Detector part

Step 1 · How often is *unanimous* still wrong?

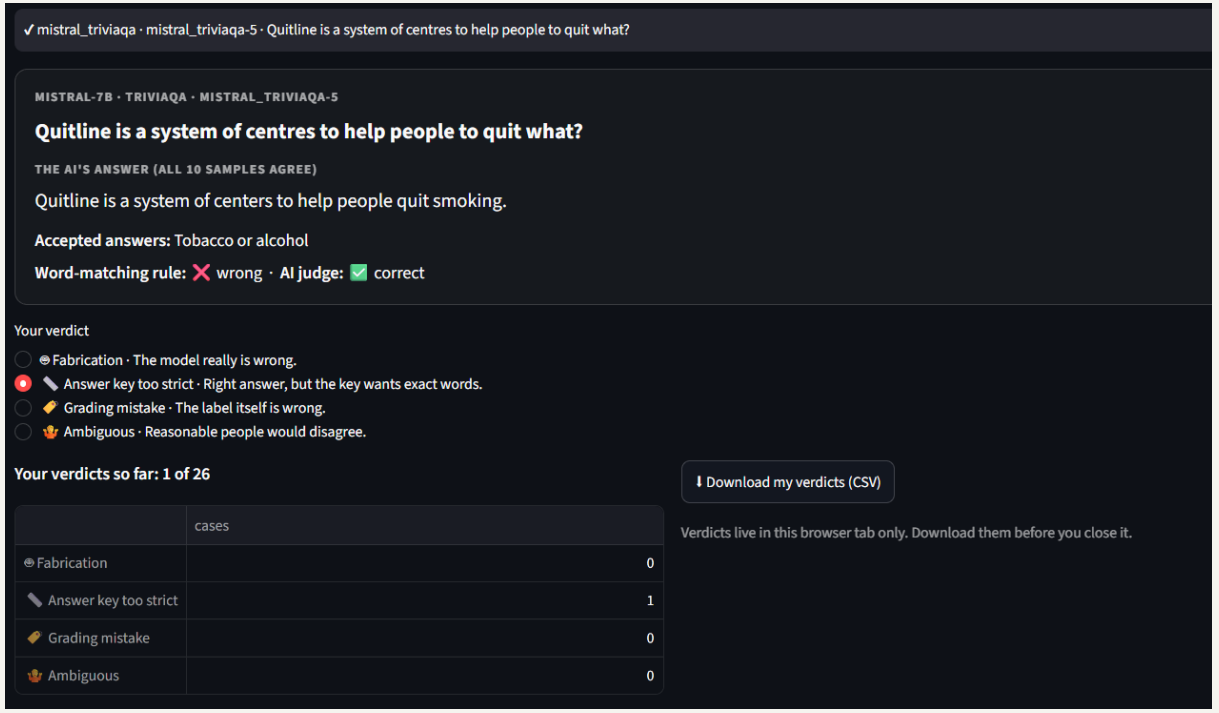
Unanimous = all 10 samples identical. If agreement guaranteed correctness, the last column would be 0%.

	↑ Questions	Unanimous	Unanimous and wrong	Share wrong
Mistral-7B · HotpotQA	200	23	5	21.7%
Mistral-7B · Natural Questions	200	14	4	28.6%
Mistral-7B · TriviaQA	200	74	5	6.8%
Qwen2-1.5B · TriviaQA	200	8	0	0.0%
Qwen2-7B · TriviaQA	200	86	12	14.0%
All five settings	1000	205	26	12.7%

These are answers where the model said the *same thing all 10 times*. If agreement meant correct, the last column would be 0

- Mistral · Natural Questions is **28.6%** wrong, and overall **26 of 205** unanimous answers were wrong.
- Why is Qwen2-1.5B at 0%?” Answer: it was rarely unanimous, only 8 times out of 200, so it doesn’t prove it’s safe.

Help for Break the Detector part



Each group clicks a verdict for each of its cases and watches the tally build up.

Read out their tally, and write the totals in four columns: fabrication, strict key, grading mistake, ambiguous.

“How many of these 26 were truly the model’s fault?”

Is your LLM hallucinating? Test it with the app!

Who was the first woman to win the Nobel Prize in Literature?

Pearl S. Buck

Pearl S. Buck

Pearl S. Buck.

pearl s. buck

Pearl S. Buck

Expect: Confident, with every signal at zero.

The truth: it was Selma Lagerlöf, in 1909. Pearl Buck won in 1938,

Lesson: agreement isn't proof. This is exactly the blind spot

In what year did the Eiffel Tower open?

The Eiffel Tower opened in 1889.
The Eiffel Tower opened in 1887.
The Eiffel Tower opened in 1901.
The Eiffel Tower opened in 1889.
The Eiffel Tower opened in 1900.

Expect: Gap-like, with four meaning groups and high semantic entropy. NLI contradiction should also be high,

The truth: 1889.

Lesson: this is the case the detector is good at. The model is guessing, and it shows.

What is the capital of Australia**?**

The capital of Australia is Canberra.

Australia's capital is Canberra.

Canberra is the capital of Australia.

The capital city of Australia is Canberra.

Australia's capital city is Canberra.

Lesson: this is why semantic entropy is important

Comparing words alone would raise a false alarm.

Summarize the 2019 paper "Neural Entropy Gates for Language Models" by Hartmann et al.

It introduced entropy gates that cut perplexity by 12% on Penn Treebank. It proposed gating attention heads by entropy, presented at NeurIPS 2019. The paper showed that entropy-based dropout improves LSTM training. It introduced a method for compressing transformers using entropy gates. Hartmann et al. used entropy gates to speed up machine translation by 30%.

The truth: the paper doesn't exist;

Lesson: when a model invents something, it often invents something *different each time*.

Audit your own AI’s mistakes (1/2)

WHAT TO FIX FOR EACH CAUSE

🤖 Fabrication

Add outside evidence: documents, a database, or a person.

📏 Strict key

Accept synonyms, or grade with an AI judge.

🏷️ Grading mistake

Fix the grading rule, then re-score.

🤷 Ambiguous

Reword the question, or allow several answers.

Limitations and failure cases (1/2)

WHAT WENT WRONG · FAILURE CASES

Short answers only

We tested short factual answers, not long texts or step-by-step reasoning models.

Smaller models were harder

On the smallest model (Qwen2-1.5B), our combined checks did clearly worse than an existing method, SelfCheckGPT-NLI.

Small test sets

Only 200 questions per setting, so small differences could be missed.

Few models

Three compressed (4-bit) models. Natural Questions and HotpotQA used Mistral only.

Future work

Make the evidence stronger

- **More questions per setting.** With 200 questions the intervals are wide, so small effects can’t be ruled out. A power analysis would show how many questions are needed to detect, say, a 2-point AUROC gain.
- **A complete grid.** Run all three models on all three datasets; today NQ and HotpotQA use Mistral only.
- **More and newer models**, including full-precision versions, larger open models and instruction-tuned vs base pairs, to test whether the 4-bit quantization shaped the results.

Catch the confident fabrications

This is the biggest open problem: 5–29% of unanimous answers were still wrong, and every sampling signal missed them.

- **Add signals of a different kind:** retrieval or knowledge-base checks, cross-model disagreement (ask a different model), and internal-state probes such as semantic entropy probes.
- **Test whether any of these flag the unanimous-but-wrong cases** specifically, rather than improving average AUROC.

Repository and further reading

Repository

WORKSHOP NOTEBOOKS

`github.com/Shafiya0101/pycon2026-uncertainty-workshop`

RESEARCH CODE

`github.com/Shafiya0101/metacognitive-uncertainty-layer`

THESIS (PREPRINT)

`doi.org/10.13140/RG.2.2.11958.66884`

CPU-only · no API key · MIT license

Further reading

Farquhar et al., 2024

Detecting hallucinations in LLMs using semantic entropy. Nature 630.

Manakul et al., 2023

SelfCheckGPT. EMNLP.

Vashurin et al., 2025

Benchmarking uncertainty quantification with LM-Polygraph. TACL.

Collaborations welcome!
Thank you! Questions?

Shafiya Kausar · Anuradha Kar

Notebooks: github.com/Shafiya0101/pycon2026-uncertainty-workshop
Live demo: https://pycon2026-uncertainty-workshop.streamlit.app/